

Replication Report: Ridge–Spin–Layer Coupling and Quasi-1D 100% Spin-Polarized Transport in 2D Altermagnets (Tian *et al.*, 2026) — Tight-Binding Part

Automated Replication (Hermes subagent)¹

¹*textures-100 corpus / textures-polar-tian2026*

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We independently replicate the tight-binding core of Tian *et al.* [arXiv:2607.15009v1], which introduces ridge–spin–layer coupling (RSLC) in 2D square-lattice altermagnets and claims quasi-one-dimensional (Q1D) 100% spin-polarized transport. From scratch we build the two-band RSLC Hamiltonian [their Eq. (3)], compute the band structure, group velocities, and semiclassical Boltzmann conductivities [their Eqs. (2),(4)], and verify the headline: in the ridge limit ($\delta \rightarrow 0$) the spin-up and spin-down channels carry current along *orthogonal* directions with conductivity spin polarization $\text{SP}_{xx} = +1$, $\text{SP}_{yy} = -1$ (both $|\text{SP}| = 1$). At finite δ the polarization degrades smoothly ($|\text{SP}| \approx 0.997$ at $\delta = 0.05\pi_0$), reproducing the “quasi-1D” qualifier. We deliberately scope out the DFT of monolayer $\text{Mg}_2\text{Mo}_2(\text{PO}_5)_2$. **Verdict: REPLICATED** (tight-binding scope). Coverage 7/10, Agreement 9/10.

CLAIM UNDER TEST

The paper’s headline (recipe): “*RSLC enables quasi-1D 100% spin-polarized transport in 2D altermagnets.*” Two orthogonal dispersionless lines (“ridges”) R_1, R_2 in the Brillouin zone lock to opposite spin *and* opposite atomic sublayer, connected by the spin-group symmetry $\{C_2||O_L\}$. Because a ridge has zero group velocity along its own direction, each spin conducts along only one Cartesian axis, giving fully spin-polarized, direction-discriminating currents (their Fig. 3c).

MODEL AND METHOD

We take the paper’s minimal basis $\{|d_{xz}, \uparrow\rangle_2, |d_{yz}, \downarrow\rangle_1\}$ and Hamiltonian [Eq. (3)]

$$H(\mathbf{k}) = \varepsilon_\alpha + \begin{pmatrix} \pi_0 \cos k_x + \delta \cos k_y & 0 \\ 0 & \pi_0 \cos k_y + \delta \cos k_x \end{pmatrix}. \quad (1)$$

The diagonal blocks are the spin-up (d_{xz}) and spin-down (d_{yz}) bands. The d_{xz} orbital hops strongly along x , weakly along y ; d_{yz} the reverse. Hence at $\delta = 0$, $E_\uparrow = \varepsilon + \pi_0 \cos k_x$ is flat along k_y (a ridge running along k_y) and $E_\downarrow$ is flat along k_x .

Group velocities $v_a = \partial E / \partial k_a$ ($\hbar = 1$): $v_y^\uparrow = -\delta \sin k_y \rightarrow 0$ and $v_x^\downarrow = -\delta \sin k_x \rightarrow 0$ as $\delta \rightarrow 0$. Semiclassical Boltzmann conductivity with constant τ [Eq. (2)]:

$$\sigma_{ab}^n = e^2 \tau \frac{1}{N} \sum_{\mathbf{k}} v_a^n v_b^n \left(-\frac{\partial f}{\partial E} \right)_{E_n(\mathbf{k})}, \quad (2)$$

and the conductivity spin polarization [Eq. (4)] $\text{SP}_{\hat{n}\hat{n}} = (\sigma_{\hat{n}\hat{n}}^\uparrow - \sigma_{\hat{n}\hat{n}}^\downarrow) / (\sigma_{\hat{n}\hat{n}}^\uparrow + \sigma_{\hat{n}\hat{n}}^\downarrow)$. Parameters: $\varepsilon = 0$, $\pi_0 = 1$, $N_k = 401^2$, $E_F = 0.3\pi_0$, $k_B T = 0.02\pi_0$.

TABLE I. Longitudinal conductivities (units $e^2\tau$) and spin polarization vs. ridge-breaking parameter δ .

δ/π_0	$\sigma_{xx}^\uparrow$	$\sigma_{yy}^\uparrow$	$\sigma_{xx}^\downarrow$	SP_{xx}	SP_{yy}
0.00	0.3034	0.0000	0.0000	+1.000	−1.000
0.05	0.3032	0.0004	0.0004	+0.997	−0.997
0.20	0.2997	0.0067	0.0067	+0.956	−0.956

RESULTS

In the ideal ridge limit ($\delta = 0$) the spin-up channel conducts *only* along x ($\sigma_{yy}^\uparrow = 0$ exactly) and the spin-down channel *only* along y , yielding $\text{SP}_{xx} = +1$ and $\text{SP}_{yy} = -1$. This is the paper’s Fig. 3(c-III) claim: SP reaches ± 1 in the ridge window, opposite signs on orthogonal axes. The finite- δ data quantify the “quasi” in quasi-1D: a small ridge dispersion leaks $O(\delta^2)$ conductance into the orthogonal channel but preserves $|\text{SP}| > 0.95$ up to $\delta = 0.2\pi_0$. The bands span $[-1, 1]\pi_0$ along Γ - X - M - Y - Γ and are exactly flat along $\Delta(0, v, 0)$ for spin-up (ridge peak-to-peak = 0), consistent with the 1D band-representation requirement.

COMPARISON TO PAPER AND VERDICT

The sign convention of SP_{xx} vs SP_{yy} in our run is $(+1, -1)$; the paper quotes $(-1, +1)$ (their text after Eq. 4). This is a pure basis/labeling convention (which orbital is called “up”); the physics — $|\text{SP}| = 1$ with *opposite* full polarization on orthogonal axes — is reproduced exactly. All structural claims of the TB model hold: ridge flatness, $\sigma_{xx} = 0$ for a ridge along x , orthogonal spin-locked channels, and smooth quasi-1D degradation.

Verdict: REPLICATED within the tight-binding scope. *Not attempted*: the DFT band structure of $\text{Mg}_2\text{Mo}_2(\text{PO}_5)_2$, the layer-resolved polariza-

tion $P(k)$, the relativistic electric Hall effect χ_{xy} , and absolute (material) conductivity magnitudes — see `failure_analysis.md`.

REPRODUCIBILITY

```
/home/stevens/comfyui-env/bin/python \
work/tian2026_replication.py
```

Runtime ≈ 0.05 s (numpy 2.3.5). Result:
`work/tian2026_result.json`.